

MSA Beyond GRR - Bias, Linearity, and Stability Studies

When and how to conduct advanced MSA studies | AIAG MSA 4th Edition

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Bias Study - Is Your Gauge Accurate?

- Bias = Average measured value - Reference (master) value
- Measures systematic error - the gauge consistently reads high or low.

Method:	Measure a reference part (certified master) 10-25 times. Calculate average. Bias = Average - Reference.
% Bias:	% Bias = (Bias / Process Variation) x 100. Should be < 10%.
t-test:	Conduct t-test to determine if bias is statistically significant. If $ t > t_{critical}$, bias is significant and must be corrected.
When Required:	At gauge first use, after calibration, after repair, as part of PPAP Element 8.
Common Cause:	Zero point error in calibration, worn reference surface, temperature effects.

Linearity Study - Is Accuracy Consistent Across Range?

- Linearity measures whether bias is consistent across the operating range of the gauge.
- A gauge may be accurate at 25mm but biased at 100mm - linearity study reveals this.

Method:	Select 5 reference parts spanning the operating range. Measure each 12 times. Calculate bias at each reference value.
Regression:	Plot bias vs reference value. Fit regression line: Bias = a + b x Reference.
R-squared:	R^2 should be > 0.95. Low R^2 means non-linear gauge behaviour.
% Linearity:	% Linearity = $ b \times \text{Process Variation} / \text{Total Variation} \times 100$
When Required:	For reference gauges, CMMs, hardness testers - any gauge used across a wide range.

Stability Study - Does the Gauge Drift Over Time?

- Stability measures whether the gauge performance changes over time.
- Uses control chart method: measure reference part 3-5 times per period, over 20+ periods.

Control Charts:	Plot X-bar (accuracy/bias trend) and R (repeatability trend) over time.
Acceptance:	All points within UCL/LCL on both charts. No trends, cycles, or out-of-control patterns.
Period:	Daily for first month, then weekly/monthly based on gauge stability history.
Action if Unstable:	Investigate cause (temperature, wear, contamination). Recalibrate. If instability recurs, consider gauge replacement.

Attribute GRR - For Go/No-Go Gauges

- Used when gauge gives a pass/fail result, not a measurement value.

Method:	3 appraisers evaluate 50 parts (mix of obvious pass, obvious fail, and borderline). Each appraiser evaluates twice (blind). Record pass/fail decisions.
Kappa Coefficient:	Kappa ≥ 0.75 : Acceptable agreement. Kappa 0.50-0.74: Marginal. Kappa < 0.50: Unacceptable.
Disagreement Analysis:	Focus on borderline parts - where appraisers disagree, there is a misclassification risk.
When Required:	For all attribute gauges in Control Plan, especially for CC/SC characteristics.